WRIGHT STATE UNIVERSITY

COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

Feature Engineering

Turning columns into better inputs for a model



What is a feature?

Features (inputs) **Target**

Hours studied	Sleep (hrs)	Major	Exam score
5	7	Bio	78
3	6	Nurs	65
8	8	Hist	91
6	5	Bio	72

Feature

An input column the model uses to learn patterns.

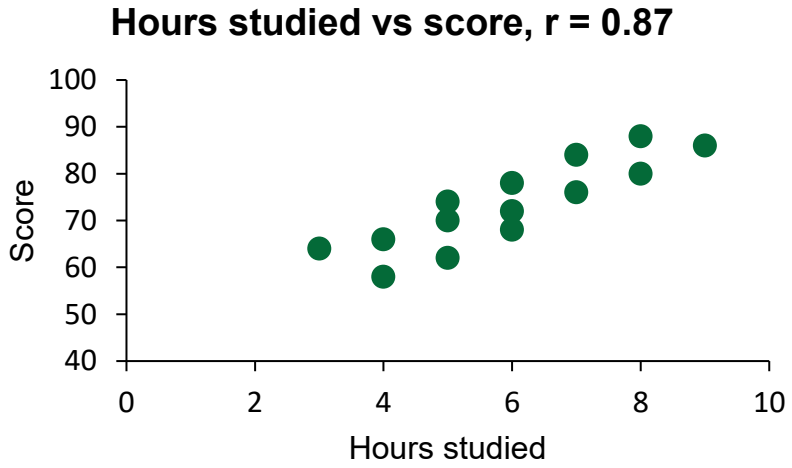
Target

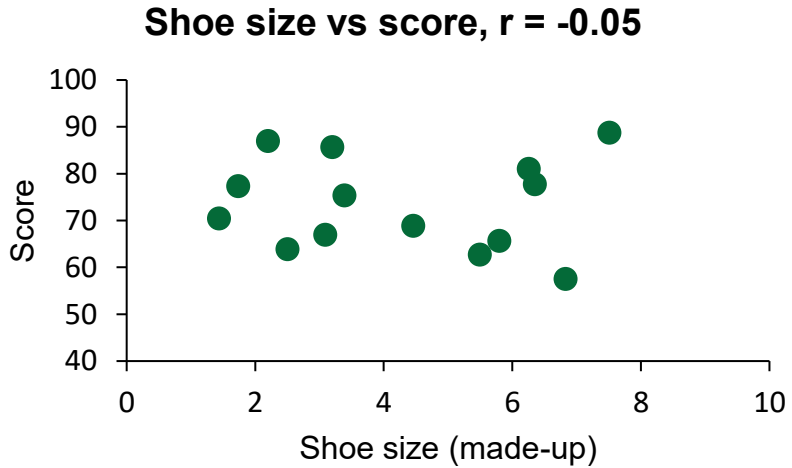
The column we want to predict.

Feature selection: keep or drop

Column	Decision	Why
Student ID	Drop	Only a label, no pattern
Name	Drop	A name does not cause a score
Hours studied	Keep	Likely linked to the score
Sleep (hrs)	Keep	Likely linked to the score
Favorite color	Drop	Irrelevant noise
Exam score	Target	What we want to predict

Spotting useless columns





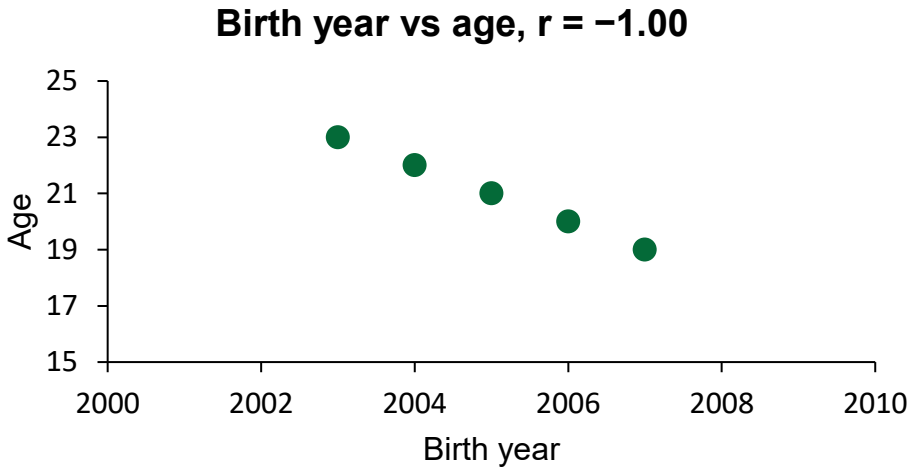
Left: keep. Right: no pattern, drop it.

Redundant columns

Birth year	Age
2006	20
2005	21
2004	22
2007	19
2003	23

Age = 2026 − Birth year.

Also redundant: weight in kg and lb, price and price with tax.



A perfect line: one column tells you the other. Keep one.

Creating new features

Beam	Strength (kN)	Weight (kg)	Strength ÷ Weight
A	400	20	20
B	600	60	10
C	300	10	30

Beam C is the best per kg, even though B is the strongest.

New feature

Combine old columns with a formula.

Needs domain knowledge

Ask what an expert would divide or multiply.

More examples of new features

Area	New feature	Formula	Example
Melt pool	Energy density	Power ÷ (Speed × Beam diameter)	200 ÷ (1000 × 0.1) = 2.0
Café	Revenue per customer	Sales ÷ Customers	$600 ÷ 150 = $4
Delivery	Average speed	Distance ÷ Time	30 km ÷ 0.5 h = 60 km/h
Dates	Day of week	From the date	2026-10-29 → Thursday
Students	Study hours per course	Hours ÷ Courses	12 ÷ 4 = 3

Encoding: words to numbers

Models only read numbers. Words must become numbers.

Label encoding

One number per category.

Small → 1

Medium → 2

Large → 3

One-hot encoding

One 0/1 column per category.

Red → 1, 0, 0

Blue → 0, 1, 0

Green → 0, 0, 1

Label encoding: use for ordered labels

Good: ordered labels

Size	Encoded
Small	1
Medium	2
Large	3
Medium	2

The order is real: 1 < 2 < 3.

Bad: unordered labels

Color	Encoded
Red	1
Blue	2
Green	3
Blue	2

Model thinks Green > Red. Wrong!

One-hot encoding: unordered labels

Color	Is Red	Is Blue	Is Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1
Blue	0	1	0

One 1 per row. No false order.

Watch out

500 cities = 500 new columns.

Fix: group rare ones as "Other".

Which encoding should I use?

Situation	Use	Example
Ordered labels	Label encoding	Small, Medium, Large → 1, 2, 3
Unordered labels	One-hot encoding	Red, Blue, Green → three 0/1 columns
Many categories	Group rare ones as “Other”, then one-hot	500 cities → top 10 + Other
Already numbers	Nothing to do	Age, price

Binning: numbers to groups

Hours studied	Group
1	Low
3	Low
5	Medium
6	Medium
8	High
12	High

Low: 0 to 3 hours

Medium: 4 to 7 hours

High: 8 or more hours

Edge: 3 hours is Low but 4 is Medium, though they are close.

Two ways to choose the bins

Equal width	1	2	2	3	5	6	9	15	40
Equal count	1	2	2	3	5	6	9	15	40

Equal width

Same size bins: 1–14, 14–27, 27–40.

Counts: 7, 1, 1 (uneven).

Equal count

Same number in each bin.

Counts: 3, 3, 3 (even). Better for skewed data.

Scaling: put numbers on the same scale

Student	Age	Income
A	20	40,000
B	50	42,000

Before scaling

Age differs by 30.

Income differs by 2,000.

Income looks 67 times bigger.

After scaling to 0–1

Student A: age 0.05, income 0.25

Student B: age 0.76, income 0.28

Age differs by 0.71.

Income differs by 0.03.

Now age is not ignored.

Min-max scaling (0 to 1)

scaled = (x − min) ÷ (max − min)

Age	Calculation	Scaled
18	(18 − 18) ÷ 42	0.00
25	(25 − 18) ÷ 42	0.17
40	(40 − 18) ÷ 42	0.52
60	(60 − 18) ÷ 42	1.00

Another way, and an outlier warning

Standardize (z-score)

z = (x − mean) ÷ std

Warning: outliers

One big value squeezes the rest.

Age	z-score
18	-0.95
25	-0.58
40	0.23
60	1.30

Hours	Min-max
3	0.00
6	0.14
9	0.27
25	1.00

Aggregation: daily to monthly

Date	Cups sold
Jan 1	120
Jan 2	135
Jan 3	150
Feb 1	100
Feb 2	90
Feb 3	110



Month	Total	Average	Max
Jan	405	135	150
Feb	300	100	110

6 daily rows become 2 monthly rows.

Watch out

Detail is lost. Pick the level that fits the question.

Aggregation: many rows per person

Customer	Amount ($)
A	5
A	12
A	8
B	20
B	4



Customer	Orders	Total	Average
A	3	25	8.33
B	2	24	12.00

5 purchase rows become 2 customer rows with new features.

Combining two data sets

Player ID	Platform
P1	PC
P2	Mobile
P3	Console

Player ID	Spend ($)
P1	12
P2	30
P4	8

Combined on Player ID (P3 and P4 are in only one table, so they are left out)

Player ID	Platform	Spend ($)
P1	PC	12
P2	Mobile	30

Merge Data in Orange

1 **Two File widgets: one table each**

2 **Connect main table to Data, second to Extra Data**

3 **Choose the key (Player ID) in both**

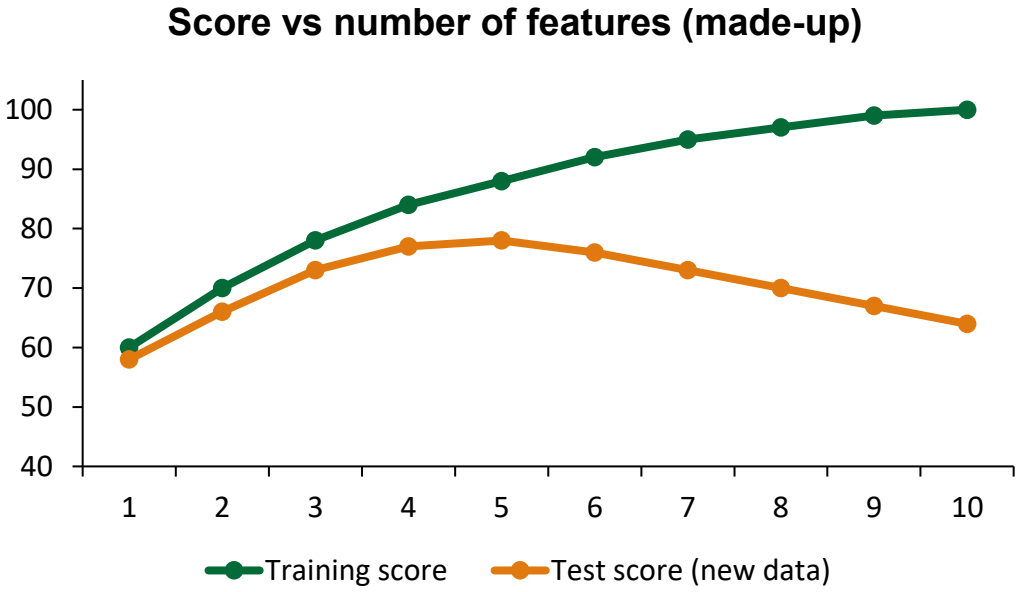
4 **Choose the merge type, then check the row count**

Merge type

Append columns: keep all main rows, blanks where no match.

Matching pairs: keep only rows found in both.

More columns is not always better



What happens

Training score keeps rising.

Test score peaks near 5 features, then falls.

Too many features cause overfitting and confuse the model.

Why too many features hurt

Noise to memorize

Useless columns let the model memorize rows instead of learning patterns.

Hard to explain

With 50 inputs nobody can say why the model decided.

Needs more rows

More columns need many more rows to learn from.

Example: 20 students, 15 columns. The model can memorize every student.

Using it on Assignment 1.3 (Melt Pool)

Technique	Idea for Melt Pool
Selection	Drop row numbers or any ID column.
Redundant	Do not use Defect Severity to predict Defect Type.
New feature	Energy density = Laser Power ÷ (Scan Speed × Beam Diameter).
Binning	Cooling Rate: Low, Medium, High. Shielding Gas: low or high.
Scaling	Laser Power, Scan Speed, Beam Diameter to 0–1.
Encoding	Defect Type labels to numbers if the model needs it.
Aggregation	Not needed: each row is already one test.

Which technique?

Situation	Technique
Column of Student ID numbers	Drop it (feature selection)
Birth Year and Age both present	Remove one (redundant)
Strength and Weight columns	Create Strength ÷ Weight (new feature)
Color: Red, Blue, Green	One-hot encoding
Hours studied → Low, Medium, High	Binning
Age 18–60 and Income 20,000–100,000	Scaling
Daily sales, question is monthly	Aggregation